

# Patient Messaging Tool — Strategy as a Game

23 September 2026 · Ian Moore · working model

Updated with Sarah's email of 22 September. Payoffs are ordinal and illustrative; trust the conclusions that do not depend on the numbers. The mathematical details are in the appendix. Not legal advice.

## What Sarah's 22 September email settled

- No information from the hospital chart or ConnectingOntario goes into the app. ConnectingOntario is for reading and gathering only; lifting information from it is not allowed. Including hospital information at this stage would be a major privacy breach.
- Patient-reported answers are in: patients answer questions themselves, which avoids the chart-privacy issue.
- v1 focus: post-op patients. A dropdown of the procedures she performs, and checkboxes rather than free-form answers, because many patients struggle to describe symptoms. Wound photo sharing would be valuable later.
- Her own clinic records (Accuro) hold the most important information; her clinic notes matter most for dealing with patients.
- The pain: post-op phone calls despite detailed written instructions. Goals: fewer calls to the office, fewer callbacks she has to make, fewer follow-up clinic visits, so she can focus on new consults and sicker patients.
- Path: a tool for her first, then her colleagues, mainly as an outpatient tool for clinic patients. If it ever goes to the hospital, the angle is preventing emergency department visits.
- She is not involving Mackenzie Health admin.

## Objective and constraint

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maximize over strategy  $\sigma$ :  
  E[  $w_1 \cdot \text{paying partners}(\sigma) + w_2 \cdot \text{capital raised}(\sigma)$  ] - Drag( $\sigma$ )  
subject to:  
  Legal( $\sigma$ )  $\geq$  Floor(scale reached)
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- **Drag decomposes by severity:** veto exposure (can end the game) is far worse than control rights ceded (IP, data, publication, timeline), which is worse than delay. Minimize in that order.
- **The legal floor is set by the success state, not today's state.** Detection probability rises with scale (patients, physicians, investor diligence), so comply to the scaled standard from day one. Compliance debt is priced by investors at diligence.
- **Product scope moves the floor.** A platform serving two or more custodians primarily to let them disclose personal health information to one another is a health information network provider under O. Reg. 329/04 s. 6, and must do and hand over a written threat, risk and privacy assessment. Privacy impact assessments are otherwise not formally required under PHIPA. Keep practices as separate tenants with no custodian-to-custodian sharing to stay out of that category; still do a light assessment as a signal.
- **Penalties are built so that shortcuts never pay.** The Information and Privacy Commissioner's administrative monetary penalties have been in force since 1 January 2024: up to \$50,000 for individuals and \$500,000 for organizations, and they can exceed those caps to strip economic benefit. The criteria include preventative steps not taken and economic benefit gained.

## Solution concepts

Moves are sequential and information is private, so plain Nash equilibrium admits non-credible threats. The model uses subgame perfection (backward induction) for sequencing, perfect Bayesian equilibrium for signaling to investors and physicians, and veto-player logic for drag.

## Players

| Player                                            | Moves                                  | Veto?                                                                                                                | Wants                                                                                                 |
|---------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ian + Sarah (founders)                            | Scope, disclosure, sequencing, signals | n/a                                                                                                                  | Partners and capital, low drag                                                                        |
| Sarah (separately)                                | Participate / exit                     | Over everything (participation constraint)                                                                           | Time back, fewer calls and follow-ups, care quality, no risk to license, privileges or board standing |
| Mackenzie admin                                   | Acknowledge / assert jurisdiction      | Only inside jurisdiction: hospital systems, data, resources, branding; her conflict of interest if she is a director | Low effort, no risk ownership, no embarrassment, good relations with medical staff leadership         |
| Ontario Health (ConnectingOntario)                | Audit, suspend or revoke access        | Out by design: no ConnectingOntario data in the app                                                                  | Use only as authorized                                                                                |
| IPC, CPSO, Health Canada                          | React to complaints or triggers        | Legal, when triggered                                                                                                | Compliance                                                                                            |
| Patients                                          | Consent, answer check-ins, message     | Over their own participation                                                                                         | Reassurance, fewer unnecessary visits                                                                 |
| Physicians                                        | Adopt and pay / not                    | Over revenue                                                                                                         | Time saved at acceptable risk and price                                                               |
| Investors (incl. IRAP)                            | Fund / pass                            | Over capital                                                                                                         | Signals they can verify                                                                               |
| Ethics board, evaluator, institutions             | Only if invited                        | Opt-in                                                                                                               | Their own process                                                                                     |
| Competitors (Epic, AI scribe vendors, SeamlessMD) | Enter / ignore                         | No                                                                                                                   | Protect or extend their footprint                                                                     |

Veto set after her email: Sarah and the law. Mackenzie admin leaves the veto set through scope design; Ontario Health leaves because the app never touches its data. Opt-in players stay out until their signal value exceeds their drag.

## Subgame 1: Mackenzie admin (disclosure)

Payoffs are (founders, admin), higher is better.  $p$  is the probability admin learns of the project other than from her. Shaded cells are admin's best response to each move.

| Founders' move                | Admin acknowledges                           | Admin asserts jurisdiction |
|-------------------------------|----------------------------------------------|----------------------------|
| Ask permission                | (2, 1): approval makes admin co-own the risk | (1, 2)                     |
| Inform, with scope boundaries | (3, 3)                                       | (2, 1)                     |
| Stay quiet, discovered        | (4, 0)                                       | (0, 2)                     |

Staying quiet and never being discovered pays the founders 4, with probability  $1 - p$ .

- Asking concedes jurisdiction, and approval transfers risk to admin, so their best response is a full review. Asking is dominated whenever admin lacks jurisdiction. This supports her decision not to involve admin.
- Asserting against a disclosed, out-of-jurisdiction private-practice tool costs admin more than acknowledging. The threat is not credible.
- **Staying quiet is a Nash equilibrium, not a subgame-perfect one.** (Quiet; admin asserts everywhere) is a Nash equilibrium for  $p \leq 1/2$ , but it rests on the off-path threat to assert after being informed. Under subgame perfection, informing (3) beats staying quiet ( $4(1 - p)$ ) once  $p > 1/4$ .
- $p$  is not fixed: success drives it toward 1 (colleagues, patients, investor diligence about her hospital role). Staying quiet is optimal only in the world where the project stays small.
- Staying quiet converts a non-credible threat into a credible one. If she is a director, an undisclosed family interest hands admin a governance hook regardless of the project's scope.
- The concern about admin is correct for anything inside their jurisdiction, which is why scope design comes first. Her email already draws that line (no hospital chart or ConnectingOntario data).

**Equilibrium:** keep the hospital out of the approval loop; if she is a director, a short written disclosure of the family interest (information, not a request); admin acknowledges.

## Subgame 2: Ontario Health (resolved by exclusion)

- Inspection-game logic: the gain from shortcuts is small and the penalty (loss of ConnectingOntario access, IPC penalties, reputation) is catastrophic, so full compliance is dominant.
- Resolved 22 September: Sarah confirms lifting information from ConnectingOntario is not allowed. Ontario Health's Acceptable Use Policy (v1.1, June 2020) likewise limits use "solely for the purposes authorized in the applicable agreement" and allows suspension or revocation.
- The chart-import stage via ConnectingOntario is dropped, and Ontario Health leaves the game. The chart-summary value moves to her own Accuro records in a later stage.

**Rule:** ask only players who already hold a veto; inform those who don't; design out the rest; invite opt-in players only when their signal is worth more than their drag.

### Subgame 3: Investors (signaling, least-cost separating)

The founders know how good the product is; investors don't. A signal separates only if a weak product cannot cheaply produce it. In the least-cost separating outcome the strong type sends the cheapest signal that clears the current investor's bar; anything more is waste, paid here in drag.

| Signal                                                                         | Can a weak product fake it?                | Drag        | Verdict                                            |
|--------------------------------------------------------------------------------|--------------------------------------------|-------------|----------------------------------------------------|
| Sarah uses it                                                                  | Yes                                        | None        | Necessary, weak signal                             |
| Her partners use it                                                            | Mostly                                     | None        | Weak                                               |
| Outside physicians pay                                                         | No                                         | None        | Strong; best ratio                                 |
| Retention over 6–12 months                                                     | No                                         | None        | Strong                                             |
| Fewer post-op calls, callbacks and follow-up visits vs pre-registered baseline | Hard, if pre-registered and tamper-evident | None        | Strong; these are the outcomes she named           |
| IRAP award                                                                     | Hard (the advisor screens)                 | Low         | Moderate                                           |
| Endorsements (Mi2, OntarioMD)                                                  | Yes; the endorser commits nothing          | High        | Cheap talk; skip                                   |
| Independent evaluation (academic + ethics board)                               | No                                         | Medium–high | Strongest; priced for Series A                     |
| Institutional buyer validation (OBIO EAHN, CAN Health)                         | No                                         | High        | Strong; hospital path only (emergency-visit angle) |

**Mechanism-design move.** Before launch, hash and anchor the metric definitions (calls to the office, callbacks, follow-up visits; later emergency visits), the analysis plan, a baseline summary and an all-users reporting commitment. The hash-chained log plus periodic anchored roots makes post-hoc metric selection and post-hoc data alteration detectable. This raises the separating power of free signals and lets the paid independent evaluator wait until the Series A bar requires it. Only hashes of plans and aggregates go on-chain.

The same log covers what PHIPA s. 10.1 would require of electronic audit logs (type of information, time, who accessed, whose record). That section was enacted in 2020 and awaits regulations, per the sources found; check its current status.

### Subgame 4: Physicians (coordination)

- Within a practice, the secretary relay disappears only if the whole practice switches: a stag hunt. Everyone adopting is payoff-dominant; no one adopting is risk-dominant when the benefit is less than twice the cost (appendix A.5). Select the good equilibrium by selling to the practice as a unit (one decision, one PHIPA agreement) with a champion. Her plan to disseminate to colleagues once it works fits this.
- Outside physicians do not adopt a tool that lacks signed agent agreements, patient consent and CMPA-consistent use. Legal compliance is a precondition of the adoption equilibrium, not just a constraint.
- A returning messaging billing code would shift physician payoffs sharply. Watch for it.

## Sarah's participation constraint

Her utility: time saved (fewer calls, callbacks, follow-ups) + upside (if equity) + care quality – professional risk – governance risk – time. Any strategy with a real chance of costing her privileges, license or board standing violates the constraint and ends the game. Her email satisfies it by construction: nothing from the hospital, patient-reported data only. Equity strengthens her commitment and the investor read (a committed clinical cofounder) but raises the weight of the conflict of interest. She frames it as “a tool meant for me at the moment”, so the venture objective and her role need explicit alignment before any investor steps.

## Competitors

AI scribe vendors have the lowest cost of entering this niche; Epic's model and buyer are elsewhere; SeamlessMD overlaps with structured post-op check-ins but sells to hospitals, not private practices. Best response: speed, switching costs (practice-level workflow, procedure and protocol library, attested history, Accuro integration), and keeping partnership or exit options open.

## Equilibrium path (backward induction)

1. Choose the game: separate entity; private-practice scope; no hospital systems, data, resources or branding; no hospital chart or ConnectingOntario data; separate tenants (not a health information network provider); red-flag rules support rather than replace her judgement.
2. If she is a director: a short written disclosure of the family interest to Mackenzie (information, not a request). Before any real patient: patient consent, a written privacy agreement between her clinic and the company, Canadian hosting.
3. IRAP and SR&ED. Measure the baseline (post-op calls, callbacks, follow-up visits); pre-register and anchor the metrics.
4. v1 with her post-op patients: procedure dropdown, checkbox check-ins, messaging.
5. Her colleagues at the practice, as a unit.
6. Outside practices, paying. Later: wound photos; Accuro notes.
7. Raise on least-cost separating signals.
8. Choose the buyer path from the data. The hospital path uses the emergency-visit angle; an independent evaluator and ethics board for the next bar; path-specific stakeholders when their triggers are met.

**Deviation check.** Admin acknowledging beats asserting; Ontario Health is out of the game; other monitors have nothing to act on; physicians adopt once it is compliant with proof and a return; investors fund on separating signals; Sarah stays because the path is safe for her. The founders would not deviate to staying quiet (dominated once p passes the threshold), to asking (concedes jurisdiction), or to early stakeholders (overpays for signals, adds veto and control exposure).

## Parameters that can flip the equilibrium

- **Her board status:** if she is a current director, disclosure is a formal conflict-of-interest obligation, not just strategy.
- **Her equity and the venture objective:** she frames the tool as hers for now; align on intent and role.
- **Free text vs structured:** she prefers patients avoid free-form answers; the current decision keeps free text alongside dropdowns. Structured-first also limits what sensitive detail enters the system.
- **Investor bar:** if Canadian health investors require institutional validation earlier, hospital-path stakeholders move up.
- **Billing code return:** shifts physician payoffs; get Ontario Health verification.
- **AI scribe vendor entry:** raises the speed premium; consider partnership.

## Appendix: mathematical details

This appendix formalises the main text. Each result is stated in general form first and with the illustrative payoffs second, so it is clear which conclusions depend on the numbers. Expected-value comparisons treat the illustrative payoffs as cardinal; the best-response results need only their ordering.

### A.1 Objective, drag and the legal floor

The founders choose a staged plan  $\sigma$  from a strategy set  $\Sigma$ . With  $n(\sigma)$  the scale reached (physicians and patients):

$$\max_{\sigma \in \Sigma} U_F(\sigma) = \mathbb{E}[w_1 N(\sigma) + w_2 K(\sigma)] - D(\sigma) \quad \text{subject to} \quad \Lambda(\sigma) \geq \bar{\Lambda}(n(\sigma))$$

$N$  is paying partners,  $K$  capital raised,  $D$  drag,  $\Lambda$  the level of legal compliance, and  $\bar{\Lambda}$  the legal floor, non-decreasing in scale. Drag sums over the other players  $j$ :

$$D(\sigma) = \sum_j \left[ v_j(\sigma) V + c_j(\sigma) + \delta_j(\sigma) \right]$$

Here  $v_j$  is the probability that  $j$  uses a veto,  $V$  the value of the whole project,  $c_j$  the control rights ceded (IP, data, publication, timeline) and  $\delta_j$  the cost of delay. Because  $V$  is everything at stake, the term  $v_j V$  dominates for any non-negligible  $v_j$ , which is the severity ordering in the main text: veto exposure first, then control, then delay.

A compliance gap that saves  $g$  and costs  $P$  if detected has expected value  $g - \pi(n)P$ , where the detection probability  $\pi(n)$  rises with scale and approaches 1 at investor diligence. The floor that binds is therefore the one at the target scale, not today's.

### A.2 Subgame 1: disclosure to hospital admin

**Moves.** The founders  $F$  choose  $A$  (ask permission),  $I$  (inform, with scope boundaries) or  $Q$  (stay quiet). After  $A$  or  $I$ , admin  $H$  chooses  $k$  (acknowledge) or  $a$  (assert jurisdiction). After  $Q$ , nature reveals the project to  $H$  with probability  $p$ ; if revealed,  $H$  chooses  $k$  or  $a$ ; if not, the game ends.

**Payoffs** are those in the Subgame 1 table. For the founders write  $q_{hi} = 4$  (quiet, never revealed),  $q_{lo} = 0$  (quiet, revealed, admin asserts),  $i = 3$  (inform, admin acknowledges),  $i_a = 2$  (inform, admin asserts) and  $\alpha = 1$  (ask, admin asserts).

**Backward induction.** Admin's best responses are  $BR_H(A) = a$  (since  $2 > 1$ ),  $BR_H(I) = k$  (since  $3 > 1$ ) and  $BR_H(Q, \text{revealed}) = a$  (since  $2 > 0$ ). The founders' continuation values are then

$$V(A) = \alpha = 1, \quad V(I) = i = 3, \quad V(Q) = (1 - p) q_{hi} + p q_{lo} = 4(1 - p).$$

**Proposition 1 (subgame-perfect equilibrium).** Informing is optimal if and only if

$$p \geq p^* = \frac{q_{hi} - i}{q_{hi} - q_{lo}} = \frac{4 - 3}{4 - 0} = \frac{1}{4}.$$

Asking is never optimal, because  $\alpha < i$ . The numerator  $q_{hi} - i$  is the cost of disclosing; when disclosure is cheap,  $p^*$  is small.

**Proposition 2 (the non-credible threat).** The profile in which the founders stay quiet and admin asserts at every information set is a Nash equilibrium whenever

$$p \leq \frac{q_{hi} - i_a}{q_{hi} - q_{lo}} = \frac{4 - 2}{4} = \frac{1}{2}.$$

Given that admin asserts everywhere,  $V(I) = 2$ , and  $V(Q) = 4(1 - p) \geq 2$  exactly when  $p \leq 1/2$ ; admin's choices after  $A$  and  $I$  are never reached, so they cost admin nothing. The profile is not subgame perfect, because after  $I$  asserting gives admin 1 rather than 3. For  $p \in (1/4, 1/2]$ , Nash equilibrium supports “stay quiet because they would block us”,

while subgame perfection prescribes informing.

**Proposition 3 (scale).** With  $p = p(n)$  increasing and  $p(n) \rightarrow 1$  at the fundraise (diligence asks about her hospital role),  $V(Q) \rightarrow q_{lo} < i$ . Staying quiet is optimal only if the project stays small.

**Robustness to an obstructive admin.** Suppose that with probability  $\theta$  admin is an obstructive type who asserts even after  $I$ . Then  $V(I) = (1 - \theta) \cdot 3 + \theta \cdot 2 = 3 - \theta$ , and

$$V(I) \geq V(Q) \iff p \geq \frac{1 + \theta}{4}, \quad V(I) \geq 2 > \alpha = V(A) \text{ for all } \theta \in [0, 1].$$

Even a certainly obstructive admin ( $\theta = 1$ ) leaves informing optimal once  $p \geq 1/2$ , and asking is still dominated.

**Robustness if she is a director.** An undisclosed family interest adds a governance cost  $\gamma > 0$  to the revealed branch, so  $q_{lo} = -\gamma$  and

$$p^* = \frac{q_{hi} - i}{q_{hi} + \gamma} = \frac{1}{4 + \gamma} < \frac{1}{4}.$$

### A.3 Subgame 2: inspection games (Ontario Health and the regulators)

The founders choose comply ( $C$ ) or shortcut ( $S$ ). A shortcut saves  $g > 0$  and costs  $P$  if detected. Detection comes from inspection, with probability  $r$  of inspecting and  $d$  of finding, and independently from complaints and diligence,  $\pi_0(n)$ :

$$\pi(r, n) = 1 - (1 - \pi_0(n))(1 - r d) \geq \pi_0(n).$$

The founders comply whatever the monitor does if and only if  $g \leq \pi_0(n) P$ . In the classic inspection game the monitor randomises, because inspecting is costly and the inspectee is left indifferent. Here the condition holds even with no inspection at all once scale makes complaints and diligence likely, so compliance is strictly dominant, the monitor never needs to inspect, and the equilibrium is pure: (comply, no inspection).

The magnitudes make this easy to satisfy.  $g$  is small (weeks of setup).  $P$  includes loss of her ConnectingOntario access, IPC administrative penalties of up to \$50,000 for an individual and \$500,000 for an organisation (more where needed to remove any benefit gained), professional discipline, and investors walking away at diligence, where  $\pi_0 \approx 1$ . Leaving ConnectingOntario data out of the product removes the shortcut action for that monitor entirely: that game has no move left to play.

### A.4 Subgame 3: signalling to investors

The product is strong ( $H$ ) or weak ( $L$ ), with prior  $\mu_0 = \Pr(H)$ . The founders send evidence  $s \geq 0$  at cost  $c_\theta(s)$ , with single crossing:  $c_L(s) - c_H(s)$  is increasing in  $s$ , so evidence is costlier for a weak product to produce. An investor in round  $r$  funds if and only if the posterior  $\mu(s) \geq \bar{\mu}_r$ , and the bar rises from round to round. Funding is worth  $B$  to the founders.

In a separating equilibrium the weak type sends nothing and the strong type sends  $s$ , which requires

$$B - c_L(s) \leq 0 \quad (\text{the weak type will not mimic}), \quad B - c_H(s) \geq 0 \quad (\text{the strong type is willing}).$$

The least-cost separating outcome, the one selected by the intuitive criterion, is

$$s^* = \min\{s : c_L(s) \geq B\}.$$

Evidence beyond  $s^*$  is waste, and here the waste is paid in drag. With several kinds of evidence  $j$ , each with separating power  $\Delta_j = c_L^j - c_H^j$  and cost to the strong type  $c_H^j + d_j$ , where  $d_j$  is the drag it brings, the efficient bundle reaches  $c_L \geq B$  at minimum total  $\sum_j (c_H^j + d_j)$ . That means favouring a high ratio  $\Delta_j / (c_H^j + d_j)$ :

- Outside physicians paying, retention and pre-registered savings have large  $\Delta_j$  and  $d_j \approx 0$ .

- Endorsements have  $\Delta_j \approx 0$ : the endorser commits nothing, so they are cheap talk.
- Independent evaluation has large  $\Delta_j$  and large  $d_j$ , so it becomes efficient only when  $\bar{\mu}_r$  rises (around Series A).

**Pre-registration as a commitment device.** Without commitment, a weak type can choose or alter metrics after the fact at some cost  $m$ , so a metric signal separates only weakly:  $c_L^{\text{metric}} \approx m$ . Anchoring a hash of the metric definitions and the analysis plan before launch, and hash-chaining the log, makes both moves detectable, which pushes the effective  $m$  towards infinity while leaving  $c_H$  unchanged. This raises  $\Delta_{\text{metric}}$  at almost no cost, the cheapest increase in separating power available. It does not stop a weak type choosing which practices to report; the all-users reporting commitment covers that.

#### A.5 Subgame 4: coordination within a practice

Two partners (the argument extends to more) choose adopt ( $A$ ) or not ( $N$ ). The secretary relay only disappears if both adopt. With benefit  $b$  and cost  $c$ , where  $c$  includes expected legal risk:

|                    | Partner 2 adopts | Partner 2 does not |
|--------------------|------------------|--------------------|
| Partner 1 adopts   | $(b - c, b - c)$ | $(-c, 0)$          |
| Partner 1 does not | $(0, -c)$        | $(0, 0)$           |

If  $b > c$ , both (adopt, adopt) and (not, not) are Nash equilibria, and (adopt, adopt) is payoff-dominant. It is also risk-dominant if and only if  $b - c > c$ , that is  $b > 2c$ ; otherwise strategic uncertainty favours nobody adopting. A single practice-level decision, with one contract and one privacy agreement, turns two choices into one and selects (adopt, adopt).

Compliance is a precondition. Write  $c = c_0 + \rho L_{\text{phys}}$ , where  $\rho$  is the probability of a legal or professional problem and  $L_{\text{phys}}$  the physician's loss if it happens. Without signed agreements and patient consent,  $\rho L_{\text{phys}}$  is large enough that  $c > b$ , and nobody adopting is the only equilibrium.

#### A.6 Sarah's participation constraint

$$u_S(\sigma) = T(\sigma) + e W(\sigma) + Q(\sigma) - \rho_{\text{prof}}(\sigma) L_{\text{prof}} - \rho_{\text{gov}}(\sigma) L_{\text{gov}} - \tau(\sigma) \geq \bar{u}_S$$

$T$  is time saved,  $e$  her equity share,  $W$  the value of the venture,  $Q$  the quality of care,  $\rho$  the probabilities of professional and governance harm with large losses  $L$  (license, privileges, board standing),  $\tau$  her time, and  $\bar{u}_S$  the status quo. Because  $L_{\text{prof}}$  and  $L_{\text{gov}}$  are large, the constraint in practice requires  $\rho_{\text{prof}} \approx \rho_{\text{gov}} \approx 0$ : it restricts which plans are feasible rather than trading off against the other terms. Her email's rules (no hospital or ConnectingOntario data) are what keep  $\rho_{\text{prof}}$  near zero for data handling. Equity raises  $e W$ , and with it her commitment and the investor read, but also the conflict-of-interest weight at an ethics board and at the hospital.

#### A.7 Veto players and when to invite a stakeholder

The project survives only if no veto player uses its veto:

$$\Pr(\text{survive}) = \prod_{j \in \mathcal{V}} (1 - v_j),$$

which falls strictly with every player added to the veto set  $\mathcal{V}$  with  $v_j > 0$ . Scope design sets  $v_H = v_O = 0$  (no jurisdiction, no data), leaving  $\mathcal{V} = \{\text{Sarah, the law}\}$ . An opt-in stakeholder  $j$  is worth inviting if and only if its marginal signal value covers its full drag:

$$\Delta B_j \geq v_j V + c_j + \delta_j.$$

Before there is usage,  $\Delta B_j \approx 0$  for most stakeholders, because there is nothing yet to evaluate or validate, so the rule excludes them. It admits the independent evaluator once the investor bar rises and there are results.

## A.8 Sequencing: irreversibility and the value of waiting

**The baseline cannot be deferred.** The savings estimate is  $\hat{\Delta} = \bar{Y}_{\text{post}} - \bar{Y}_{\text{pre}}$ , where  $\bar{Y}$  is the average of post-op calls, callbacks and follow-up visits per patient. With a comparator surgeon  $C$ , the difference-in-differences estimate removes trends common to both:

$$\hat{\Delta}_{\text{DiD}} = (\bar{Y}_{\text{post}}^T - \bar{Y}_{\text{pre}}^T) - (\bar{Y}_{\text{post}}^C - \bar{Y}_{\text{pre}}^C).$$

$\bar{Y}_{\text{pre}}$  can only be observed before launch, so skipping the baseline destroys the estimate for good. It is the one move in the plan that cannot be postponed.

**The buyer path should wait for data.** Deferring the choice between selling to practices ( $V_A$ ) and to institutions ( $V_B$ ) until the first round of data arrives is worth

$$\mathbb{E}[\max(V_A, V_B)] - \max(\mathbb{E}[V_A], \mathbb{E}[V_B]) \geq 0$$

by the convexity of the maximum (Jensen's inequality). That is why the path commits to neither before the data are in.

## A.9 What is robust and what is not

**Robust to the numbers:** admin's best responses (acknowledge an out-of-jurisdiction disclosure, review a request for approval) and hence informing over asking; compliance dominance at scale; least-cost signalling and the value of pre-registration; survival falling with each veto player; the value of waiting on the buyer path.

**Dependent on parameters:** the threshold  $p^*$ , the share  $\theta$  of obstructive admin, whether  $b > 2c$  within a practice, and the investor bars  $\bar{\mu}_r$ .

## Sources

- Information and Privacy Commissioner of Ontario, [Privacy Impact Assessment Guidelines for PHIPA](#) (health information network provider requirements, O. Reg. 329/04 s. 6)
- Blakes, [Update on Ontario's PHIPA: Administrative Monetary Penalties Now in Force](#)
- Fasken, [Significant Changes to Ontario's Health Privacy Law: Technology Providers Take Note](#) (2020; electronic audit logs)
- Ontario Health, [Acceptable Use Policy](#), v1.1, June 2020
- Ontario Health, [Secure Messaging Proof-of-Concept Pilot](#) (ended 31 March 2026)